

Quality Management System

Extended Assessment and Execution Plan

Assumption: SAP CDS/OData views are built and provide all upstream PO, BOL, container, shipment, vendor, and material data. SQL Server owns QMS data from Customer Arrival onward. The solution does not interfere with SAP standard QM.

Primary recommendation

Build the system as an **external quality execution and reporting application**. SAP remains the upstream system of record for PO, BOL, container, vendor, shipment, material, plant, storage location, quantities, and master classification attributes. SQL Server becomes the system of record only from the **Customer Arrival / Arrival Overview** stage forward.

1. Reconfirmed Business Scope

The source process begins after SAP purchasing and logistics work is already performed. Purchase orders are created in SAP S/4HANA for imported fruits and vegetables; logistics updates shipment details, especially BOL number; receiving splits quantities by container; and then the quality process starts by creating a preliminary inspection called Arrival Overview. The input also states that the user searches by container and, if the same container appears under older BOLs, the system must force BOL selection to avoid inspecting the wrong container.

The core functional chain is:

- SAP S/4HANA creates and maintains PO, vendor, material, BOL, shipment, and container-related data.
- External QMS reads these details through CDS/OData.
- User creates Customer Arrival / Arrival Overview in QMS.
- QMS generates an internal shipment or arrival reference for downstream navigation.
- User opens a quality order for one arrival and selected materials.
- User creates one or more samples per material.
- Each sample captures sample attributes, readings, observations, and defects.
- User uploads arrival and inspection images.
- User closes quality order.
- System generates PDF quality control report with shipment data, material data, samples, defects, calculations, and image thumbnails.

2. Corrected System Boundary

In scope for SAP CDS/OData

- Purchase order header and item data.
- BOL number.
- Container number.
- Vendor/supplier identity and name.
- Material number, material description, material group, material category, variety, origin, class, brand, pack type, net weight, material size, and UoM where available.
- Plant, storage location, quantity, batch, and receiving context.
- Shipment attributes such as loading date, sailing date, examination date, arrival date, transit days, loading country, loading port, arrival place, vessel name, voyage number, transport mode, truck number/type, seal number, estimated arrival date, and other logistics attributes exposed by CDS.

In scope for SQL Server

- User-selected arrival instance.
- Snapshot of SAP data used at arrival creation time.
- Arrival checklist answers.
- Quality order header and lifecycle.
- Quality order material rows.
- Samples.
- Sample readings.
- Sample observations.
- Defects captured per sample.
- Material-size override, approval, and reason.
- Images and thumbnails.
- Report generation logs.
- Status history.
- Audit trail.
- System configuration.

Explicitly out of scope

- No posting to SAP QM inspection lots.
- No SAP usage decision.
- No change to SAP material master from this application.
- No direct SQL read against SAP tables.
- No custom update of SAP standard purchasing/logistics tables.

3. Deep Assessment of Current Design

3.1 Strong points

- The process starts at the right operational point: after logistics/receiving have enough container and BOL data.
- The user requirement correctly identifies container ambiguity across BOLs.
- The design separates arrival checklist from quality order, which is good because not every arrival must immediately become a completed inspection.
- The report concept is clear: shipment details, material details, sample summary, defects, and images.
- The image requirement is practical: arrival images and quality inspection images must be separate.
- Admin-controlled thumbnail sizing is useful for both screen performance and PDF report layout.

3.2 Main gaps to fix

Area	Gap	Required correction
Data ownership	Earlier design mixed SAP upstream data and QMS transactional data.	Treat SAP CDS/OData as upstream read model; SQL Server stores QMS execution from arrival onward.
Normalization	Defects are currently represented as fixed fields in line tables.	Store defects as rows in sample_defect, linked to defect_catalog.
Sample model	Sample attributes, readings, observations, and defects are mixed.	Split sample header, reading rows, observation rows, defect rows, and optional sample image links.
Container identity	Container alone is not unique.	Use container_no + bol_no + purchase_order + purchase_order_item + material_no + batch/plant where applicable.
Material size	Override exists but lacks governance.	Add override flag, reason, old value, new value, approver, timestamp, audit row.

Area	Gap	Required correction
Images	Image flags exist as CHAR fields, but image metadata is not normalized.	Store image assets separately and link them to arrival, quality order, material, or sample.
Status	Status text exists but no controlled lifecycle.	Use status codes, status history, allowed transitions, and editability matrix.
Report	PDF expected but calculation logic is not formalized.	Store report snapshot and calculation version to make reports reproducible.
Concurrency	Shipment number/sample number generation can collide.	Use SQL sequences/identity and unique filtered indexes.

4. Target Architecture

4.1 Components

- ****Frontend Web App****

- Arrival search and arrival checklist.
- Quality order cockpit.
- Material tab.
- Sample entry panel.
- Dynamic defect panels by material group.
- Image gallery with thumbnails.
- PDF preview and generation.
- Admin screens for configuration.

- ****Backend API****

- .NET Core Web API is recommended because it integrates strongly with SQL Server, background jobs, authentication, and PDF/image pipelines.
- Responsible for SAP OData consumption, QMS transactions, validation, file upload, report generation, and audit.

- ****SAP Integration Layer****

- Reads already-built CDS Views via OData.
- Provides search by container, BOL, PO, material, vendor, and shipment.
- Handles pagination, filtering, retries, token renewal, error mapping, and snapshot creation.

- ****SQL Server Database****

- Stores QMS-owned transactions and SAP snapshots at time of arrival/order creation.
- Enforces keys, foreign keys, status constraints, audit, and soft delete rules.

- ****File/Object Storage****

- Stores original images and derived thumbnails.
- SQL Server stores metadata, ownership, display sequence, checksum, upload user, and usage scope.

- ****Background Workers****

- Thumbnail generation.
- PDF generation.
- SAP material cache refresh if needed.
- Cleanup of orphaned uploads.
- Report archive generation.

5. SAP CDS/OData Read Strategy

5.1 Assumption

All CDS Views already exist and are published as OData services. The application must act as a consumer only.

5.2 Required OData read models

Logical read model	Purpose	Minimum fields
Container search	Find containers and disambiguate by BOL/PO.	container_no, bol_no, ebeln, ebelp, material_no, quantity, plant, storage_location, batch, vendor, vendor_name
Arrival shipment details	Populate arrival and shipment snapshot.	vessel, voyage, carrier, loading/sailing/examination/arrival/unloading dates, transit days, loading port/country, arrival place, seal, truck, transport mode
Material details	Populate material tab and sample defaults.	material_no, description, group, group description, major category, origin, variety, class, net weight, material size, brand, pack type, UoM
Quantity and item context	Validate received quantity and material rows.	ebeln, ebelp, menge, meins, batch, plant, storage location
Vendor/supplier	Report display and filtering.	supplier id, name, country, optional region

5.3 Search behavior

Search must support these scenarios:

- User enters only ****container number****.
 - If exactly one active SAP row is found, show it.
 - If multiple rows exist across BOLs or POs, force user to select BOL/PO.
 - If no row is found, show a controlled error and allow search by BOL or PO.
- User enters ****container + BOL****.
 - Return rows for that exact pair.
 - If more than one PO/material row exists, show all rows in the grid.
- User enters ****PO****.
 - Return all related container/material rows for inspection selection.
- User opens existing Arrival Overview.
 - Read from SQL snapshot, not SAP, unless user presses an authorized refresh action before quality order opening.

5.4 Snapshot rule

At the moment Arrival Overview is saved, write a **SAP data snapshot** into SQL Server. This is not master-data ownership; it is an audit snapshot showing exactly what the inspector saw when the process started.

Recommended rule:

- Before arrival save: data is live from SAP OData.
- After arrival save: data is read from SQL snapshot.
- Authorized refresh allowed only while arrival status is Draft and no quality order has been opened.
- Once quality order is Open or Closed, snapshot is locked.

6. Corrected Normalized SQL Server Design

The current spreadsheet has four broad groups: Arrival, Shipment, Header, and Lines. The design should be normalized into the following tables.

6.1 Core identity tables

`qms\_arrival`

Stores the Customer Arrival / Arrival Overview header.

Column	Type suggestion	Notes
arrival_id	bigint identity PK	Internal technical key.
arrival_no	varchar(20) unique	User-facing number.
source_system	varchar(20)	Example: S4HANA.
bol_no	varchar(35)	From SAP OData snapshot.
container_no	varchar(35)	From SAP OData snapshot.
ebeln	varchar(10)	SAP PO number.
bukrs	varchar(4)	Company code.
vendor_no	varchar(10)	Supplier.
vendor_name	nvarchar(80)	Snapshot.
status_code	varchar(20)	Draft, Completed, Cancelled.
created_at	datetime2	Use UTC.
created_by	nvarchar(80)	Application user.
completed_at	datetime2 null	Set when arrival checklist completed.
row_version	rowversion	Optimistic locking.

`qms\_arrival\_sap\_snapshot`

Stores exact SAP/OData values used at arrival save.

Column	Type suggestion	Notes
snapshot_id	bigint identity PK	
arrival_id	bigint FK	Parent arrival.
odata_service_name	nvarchar(120)	Which CDS/OData service returned the data.
odata_query_hash	char(64)	Hash of query/filter.
payload_json	nvarchar(max)	Full immutable source payload.
captured_at	datetime2	UTC.
captured_by	nvarchar(80)	User or system.

`qms\_arrival\_item`

One row per SAP material/item/container line selected for arrival.

Column	Type suggestion	Notes
arrival_item_id	bigint identity PK	
arrival_id	bigint FK	
ebeln	varchar(10)	PO.
ebelp	varchar(5)	PO item.
material_no	varchar(40)	MATNR compatible.
material_desc	nvarchar(120)	Snapshot.
plant	varchar(4)	WERKS.
storage_location	varchar(4)	LGORT.
batch_no	varchar(20) null	CHARG.
quantity	decimal(18,3)	SAP quantity.
uom	varchar(3)	MEINS.
material_group	varchar(9)	MATKL.
material_group_desc	nvarchar(80)	Snapshot.
major_category	nvarchar(80)	Example: Apples.
row_version	rowversion	

Unique index: (arrival_id, ebeln, ebelp, material_no, plant, storage_location, batch_no).

6.2 Arrival checklist tables

`qms\_arrival\_checklist`

Stores checklist-level answers, not image files.

Column	Type suggestion	Notes
checklist_id	bigint identity PK	
arrival_id	bigint unique FK	One checklist per arrival.
seal_no	varchar(30) null	From SAP or user check.
carrier_name	nvarchar(80) null	Snapshot or user-entered.
seal_intact	bit null	External inspection.
seal_matches_documents	bit null	External inspection.
external_damage_exists	bit null	External inspection.
set_temperature	decimal(6,2) null	Temperature unit reading.
display_temperature	decimal(6,2) null	Temperature unit reading.
cargo_smell_normal	bit null	Internal inspection.
visual_cargo_acceptable	bit null	Internal inspection.
cargo_shifted_collapsed_water	bit null	Internal inspection.
pulp_temp_front	decimal(6,2) null	Current workbook has multiple temp fields.
pulp_temp_middle	decimal(6,2) null	Normalize spelling.
pulp_temp_back	decimal(6,2) null	
data_logger_located	bit null	
data_logger_serial	nvarchar(50) null	
data_logger_photo_taken	bit null	Flag only; actual image in image table.
logger_handed_over	bit null	
logger_active_data_available	bit null	
logger_temperature	decimal(6,2) null	
notes	nvarchar(max) null	Replaces generic logger_note when needed.

`qms\_shipment\_snapshot`

Stores shipment details shown in the quality order shipment tab and report.

Column	Type suggestion	Notes
shipment_snapshot_id	bigint identity PK	
arrival_id	bigint unique FK	One shipment snapshot per arrival.
internal_shipment_no	varchar(20) unique	Generated by QMS if needed.
loading_date	date null	
sailing_date	date null	
examination_date	date null	
arrival_date	date null	
unloading_date	date null	
inspection_date	date null	
transit_days	smallint null	
time_bar	smallint null	
loading_port	nvarchar(60) null	
loading_country	nvarchar(60) null	
arrival_place	nvarchar(80) null	
vessel_name	nvarchar(80) null	
voyage_number	nvarchar(50) null	
pullout_date	date null	
receive_date	date null	
time_bar_exceeded	bit null	
inspection_point	nvarchar(60) null	
joint_survey	bit null	
status_code	varchar(20)	Draft/Confirmed.

6.3 Quality order tables

`qms\_quality\_order`

Header table for a quality order created after Arrival Overview completion.

Column	Type suggestion	Notes
quality_order_id	bigint identity PK	
quality_order_no	varchar(20) unique	User-facing number, replaces QCNO.
arrival_id	bigint FK	Quality order requires completed arrival.
status_code	varchar(20)	Initial, Open, Closed, Reopened, Cancelled.
opened_at	datetime2 null	Set by Open Quality Order.
opened_by	nvarchar(80) null	
closed_at	datetime2 null	Set by Close Quality Order.
closed_by	nvarchar(80) null	
close_reason	nvarchar(500) null	
reopened_at	datetime2 null	Last reopen. Full history is in status table.
reopened_by	nvarchar(80) null	Site Admin only.
reopen_reason	nvarchar(500) null	Mandatory.
row_version	rowversion	

Constraint: only one active quality order per arrival unless business later approves multi-order scenarios.

`qms\_quality\_order\_material`

Materials selected inside a quality order.

Column	Type suggestion	Notes
qo_material_id	bigint identity PK	
quality_order_id	bigint FK	
arrival_item_id	bigint FK	Material must come from arrival item.
material_no	varchar(40)	Snapshot copy for performance/report.
material_desc	nvarchar(120)	Snapshot.
origin	nvarchar(60) null	
variety	nvarchar(80) null	
class	nvarchar(80) null	Avoid reserved word in SQL; use material_class if preferred.
net_weight	decimal(18,3) null	
material_size	nvarchar(20) null	Carton count/size.
material_group	varchar(9)	
material_group_desc	nvarchar(80)	
major_category	nvarchar(80)	
brand	nvarchar(80) null	
pack_type	nvarchar(80) null	
size_overridden	bit default 0	
original_material_size	nvarchar(20) null	
override_material_size	nvarchar(20) null	
override_reason	nvarchar(500) null	Mandatory if overridden.
override_approved_by	nvarchar(80) null	Optional approval.
override_approved_at	datetime2 null	

6.4 Sample normalization

The sample should not be one wide table containing all possible defects and readings. A sample is a parent record with several child collections.

`qms\_sample`

One inspection sample for one quality order material.

Column	Type suggestion	Notes
sample_id	bigint identity PK	
quality_order_id	bigint FK	Useful for fast query.

Column	Type suggestion	Notes
qo_material_id	bigint FK	Parent material.
sample_no	int	Sequential within material or order.
carton_count	smallint null	Current workbook field.
carton_identifier	nvarchar(50) null	If inspecting specific carton(s).
sample_scope	varchar(20)	OneCarton, MultiCarton, Pallet, Lot.
sample_size	smallint null	Default from material size or manual.
grower	nvarchar(80) null	
pallet_no	nvarchar(50) null	
grower_pallet	nvarchar(50) null	Current field should be clarified.
pack_code	nvarchar(80) null	
date_code	nvarchar(80) null	
label_value	nvarchar(80) null	Current LABEL spelling corrected.
lot_no	nvarchar(80) null	
packaging_material	nvarchar(80) null	
created_at	datetime2	
created_by	nvarchar(80)	
updated_at	datetime2 null	
updated_by	nvarchar(80) null	
is_deleted	bit default 0	Soft delete.
deleted_at	datetime2 null	
deleted_by	nvarchar(80) null	
row_version	rowversion	

Unique index: (qo_material_id, sample_no) where is_deleted = 0.

`qms\_sample\_reading`

Generic numeric or text readings per sample.

Column	Type suggestion	Notes
reading_id	bigint identity PK	
sample_id	bigint FK	
reading_type_code	varchar(40)	BRIX, PUC, PHC, STICKER, FIRMNESS, GROSS_WEIGHT, NET_WEIGHT, TARA, COLOUR, WAXING, DOWNGRADE.
numeric_value	decimal(18,4) null	For measurable readings.
text_value	nvarchar(100) null	For GOOD/BAD or descriptive readings.
unit_code	varchar(20) null	Kg, %, count, etc.
is_within_spec	bit null	Calculated or entered.
reading_sequence	int	Display order.
created_at	datetime2	
created_by	nvarchar(80)	

This table fixes the workbook problem where readings such as BRIX, PUC, PHC, STICKER, FIRMNESS, WAXING, DOWNGRADE, GROSS_WEIGHT, NET_WEIGHT, and TARA were placed as fixed columns in Lines.

`qms\_sample\_observation`

Captures narrative or coded observations that are not defects.

Column	Type suggestion	Notes
observation_id	bigint identity PK	
sample_id	bigint FK	
observation_type_code	varchar(40)	Appearance, Color, Packaging, Temperature, InspectorNote, Other.
observation_text	nvarchar(max)	
severity_code	varchar(20) null	Info, Warning, Critical.
created_at	datetime2	
created_by	nvarchar(80)	

`qms\_defect\_catalog`

Master catalog of all possible defects.

Column	Type suggestion	Notes
defect_id	int identity PK	
defect_code	varchar(50) unique	WASTE_DECAY, SCALD, BRUISING, CRACK, etc.
defect_name	nvarchar(100)	Display name.
defect_category	varchar(20)	Major, Minor, Critical, Other.
default_unit	varchar(20)	Count, %, Score.
is_active	bit	
sort_order	int	UI/report order.

`qms\_material\_group\_defect`

Controls which defects appear for which material group/category.

Column	Type suggestion	Notes
material_group_defect_id	int identity PK	
material_group	varchar(9)	From SAP material group.
major_category	nvarchar(80) null	Optional if material group is too broad.
defect_id	int FK	
is_required	bit	Whether UI must show/capture.
is_active	bit	
display_section	varchar(20)	Major, Minor, Readings.
sort_order	int	

`qms\_sample\_defect`

Stores actual defect values per sample.

Column	Type suggestion	Notes
sample_defect_id	bigint identity PK	
sample_id	bigint FK	
defect_id	int FK	
defect_value	decimal(18,4) null	Count/score/percentage.
defect_percentage	decimal(9,4) null	Calculated using sample size or carton count.
severity_code	varchar(20)	Major/Minor/Critical inherited but overridable if allowed.
comment	nvarchar(500) null	
is_within_tolerance	bit null	Calculated.
created_at	datetime2	
created_by	nvarchar(80)	

Unique index: (sample_id, defect_id) unless duplicate observations per defect are intentionally allowed.

6.5 Image model

`qms\_image\_asset`

Stores the physical image metadata once.

Column	Type suggestion	Notes
image_id	bigint identity PK	
storage_provider	varchar(30)	Local, AzureBlob, AWS_S3.
original_file_name	nvarchar(255)	
content_type	varchar(100)	image/jpeg, image/png.
file_size_bytes	bigint	
storage_url	nvarchar(1000)	Private object key or URL.
thumbnail_url	nvarchar(1000) null	Derived thumbnail.
checksum_sha256	char(64)	Duplicate detection/integrity.
width_px	int null	

Column	Type suggestion	Notes
height_px	int null	
uploaded_at	datetime2	
uploaded_by	nvarchar(80)	
is_deleted	bit default 0	Soft delete.

`qms\_image\_link`

Links one image to a business object.

Column	Type suggestion	Notes
image_link_id	bigint identity PK	
image_id	bigint FK	
owner_type	varchar(30)	Arrival, ArrivalChecklist, QualityOrder, QualityOrderMaterial, Sample.
owner_id	bigint	ID of owner.
image_category	varchar(40)	ArrivalExternal, ArrivalInternal, TemperatureDisplay, DataLogger, QualityInspection, SampleDefect, ReportOnly.
display_order	int	
caption	nvarchar(255) null	
include_in_report	bit default 1	
created_at	datetime2	
created_by	nvarchar(80)	

This replaces image URL text fields and photo flags as the actual image store. Existing checklist flags remain only as yes/no checklist answers.

6.6 Status, audit, and configuration

`qms\_status\_history`

Column	Type suggestion	Notes
status_history_id	bigint identity PK	
entity_type	varchar(40)	Arrival, QualityOrder, Sample.
entity_id	bigint	
old_status	varchar(30) null	
new_status	varchar(30)	
reason	nvarchar(500) null	Mandatory for close/reopen/cancel.
changed_at	datetime2	
changed_by	nvarchar(80)	

`qms\_audit\_log`

Column	Type suggestion	Notes
audit_id	bigint identity PK	
entity_type	varchar(40)	
entity_id	bigint	
action_code	varchar(40)	Insert, Update, Delete, Open, Close, Reopen, Override.
old_values_json	nvarchar(max) null	
new_values_json	nvarchar(max) null	
changed_at	datetime2	
changed_by	nvarchar(80)	
source_ip	varchar(45) null	

`qms\_system\_config`

Column	Type suggestion	Notes
config_key	varchar(100) PK	Example: thumbnail_size_screen.
config_value	nvarchar(500)	

Column	Type suggestion	Notes
data_type	varchar(20)	Int, Decimal, String, Bool, Json.
updated_at	datetime2	
updated_by	nvarchar(80)	

7. Status Lifecycle and Editability Matrix

7.1 Arrival status

Status	Meaning	Allowed actions
Draft	SAP data selected, checklist not complete.	Edit checklist, refresh SAP snapshot, upload arrival images, cancel.
Completed	Checklist complete and locked for quality order.	Create quality order, view, report arrival checklist.
Cancelled	Arrival abandoned.	View only.

7.2 Quality order status

Status	Meaning	Allowed actions
Initial	Quality order created but not opened.	Open, view, cancel if no samples.
Open	Active inspection.	Add/edit/delete samples, readings, observations, defects, images.
Closed	Finished.	View/report only. No edits.
Reopened	Closed order reopened by Site Admin.	Limited edit depending on reason.
Cancelled	Invalid order.	View only.

7.3 Editability rules

- Arrival must be **Completed** before creating a quality order.
- Quality order must be **Open** before creating samples.
- Sample save is blocked if required fields for the material group are missing.
- Closed quality orders are read-only.
- Reopen requires Site Admin role and mandatory reason.
- Material size override requires reason and audit.
- Sample delete should be soft delete only.
- Image delete should be soft delete, not physical delete, until retention job runs.

8. API Execution Plan

8.1 SAP integration APIs

API	Method	Purpose
/api/sap/containers/search	GET	Search by container, BOL, PO.
/api/sap/containers/{selectionKey}	GET	Get selected SAP container/material details.
/api/sap/materials/{materialNo}	GET	Get material details.
/api/sap/health	GET	Check OData availability.

8.2 Arrival APIs

API	Method	Purpose
/api/arrivals	POST	Create arrival from selected SAP OData rows.
/api/arrivals/{id}	GET	Get arrival, checklist, shipment snapshot, items, images.
/api/arrivals/{id}/checklist	PUT	Save checklist.

API	Method	Purpose
/api/arrivals/{id}/complete	POST	Complete arrival.
/api/arrivals/{id}/images	POST	Upload arrival images.
/api/arrivals/{id}/report	GET	Generate arrival checklist report.

8.3 Quality order APIs

API	Method	Purpose
/api/quality-orders	POST	Create quality order from completed arrival.
/api/quality-orders/{id}	GET	Full quality order cockpit.
/api/quality-orders/{id}/open	POST	Open order.
/api/quality-orders/{id}/close	POST	Close order.
/api/quality-orders/{id}/reopen	POST	Site Admin only.
/api/quality-orders/{id}/materials	GET	Materials in order.
/api/quality-orders/{id}/report	POST	Generate PDF report.

8.4 Sample APIs

API	Method	Purpose
/api/qomaterials/{qoMaterialId}/samples	POST	Create sample.
/api/samples/{id}	PUT	Update sample header.
/api/samples/{id}/readings	PUT	Replace/save readings.
/api/samples/{id}/observations	POST	Add observation.
/api/samples/{id}/defects	PUT	Save defect values.
/api/samples/{id}/images	POST	Upload sample/inspection images.
/api/samples/{id}	DELETE	Soft delete sample.

8.5 Admin APIs

API	Method	Purpose
/api/admin/defects	CRUD	Maintain defect catalog.
/api/admin/material-groups/{group}/defects	CRUD	Link defects to material group.
/api/admin/readings	CRUD	Maintain reading types.
/api/admin/config	GET/PUT	Thumbnail sizes, report settings, material sync schedule.
/api/admin/audit	GET	Audit query.

9. Detailed Execution Plan

Phase 0 - Alignment and Freeze Decisions

Duration: 3-5 working days

Decisions to finalize:

- Confirm that SAP CDS/OData views are read-only and available.
- Confirm which fields are mandatory for arrival creation.
- Confirm whether one arrival can have one quality order only.
- Confirm whether one sample is one carton, multi-carton, pallet, or configurable.
- Confirm exact defect formulas and acceptance thresholds.
- Confirm image storage provider.
- Confirm roles: Standard user, Quality Inspector, Admin, Site Admin.

Deliverables:

- Signed field mapping from SAP OData to QMS arrival snapshot.
- Approved SQL Server normalized ERD.
- Approved status lifecycle.
- Approved report mockup version 1.

Phase 1 - Database Foundation

Duration: 1-2 weeks

Tasks:

- Create SQL Server schema `qms`.
- Create identity/sequences for `arrival\_no`, `quality\_order\_no`, and sample numbering.
- Create normalized tables listed in section 6.
- Add FK constraints with `ON DELETE NO ACTION` or restrict behavior.
- Add filtered unique indexes for active records.
- Add rowversion columns for concurrency.
- Add audit triggers or application-level audit interceptor.
- Seed defect catalog for apple defects currently shown in the screen:
 - WASTE_DECAY, SCALD, BITTER_PIT_PLARA, LENTICELS_BREAKDOWN.
 - STEM_INJURY, BRUISING, CRACK, RUSSETING, SHRIVELLING, SUNBURN, WAX_RESIDUE, RED_BLUSH, INSECT_DAMAGE, CHEMICAL_RESIDUE, CALYX_MOLDS, MECHANICAL_INJURY.
- Seed reading types:
 - BRIX, PUC, PHC, STICKER, FIRMNESS, GROSS_WEIGHT, NET_WEIGHT, TARA, COLOUR, WAXING, DOWNGRADE, PACKAGING_MATERIAL.
- Seed system configuration:
 - thumbnail_size_screen_height.
 - thumbnail_size_screen_width.
 - thumbnail_size_pdf_height.
 - thumbnail_size_pdf_width.
 - image_fit_mode: Cover/Contain.

Acceptance criteria:

- DB deployment script runs repeatedly in dev.
- Foreign keys prevent orphan samples, defects, and images.
- Defects are rows, not columns.
- Sample readings are rows, not columns.
- Quality order cannot exist without arrival.
- Sample cannot exist without quality order material.

Phase 2 - SAP OData Consumer Layer

Duration: 1-2 weeks

Tasks:

- Implement typed client for each SAP OData read model.
- Implement request filtering:

- container number.
- BOL number.
- PO number.
- material number.
 - Implement container ambiguity handling.
 - Implement retry policy:
 - 3 retries.
 - exponential backoff.
 - no retry on authorization errors.
 - Implement OData error mapper.
 - Implement response hash to detect if SAP payload changed before arrival save.
 - Implement optional memory/Redis cache for search results before save only.
 - Store raw OData payload JSON when creating arrival snapshot.

Acceptance criteria:

- Searching by container returns all possible BOL/PO combinations.
- User cannot proceed when container is ambiguous until BOL/PO is selected.
- Arrival save stores both normalized columns and raw payload JSON.
- No direct SQL read from SAP is used.

Phase 3 - Arrival Overview Module

Duration: 2 weeks

Tasks:

- Build arrival search screen.
- Build SAP result grid.
- Build arrival checklist form sections:
 - Arrival header details.
 - External inspection.
 - Temperature unit reading.
 - Internal inspection.
 - Pulp temperature check.
 - Data logger inspection.
 - Notes/observations.
 - Attachments checklist.
 - Implement save draft.
 - Implement complete arrival.
 - Implement image upload for arrival categories.
 - Implement arrival report preview.
 - Implement validation before completion.

Key validations:

- Container, BOL, PO, and at least one material row are required.
- If `external\_damage\_exists = true`, an external damage note or image is required.

- If data logger is located, serial number should be required.
- If temperature display photo flag is true, at least one matching image link should exist.
- Arrival cannot be completed if mandatory checklist answers are missing.

Acceptance criteria:

- Arrival Overview can be created from SAP OData rows.
- Completed Arrival cannot be changed except by authorized correction workflow.
- Arrival images are separated from quality order inspection images.

Phase 4 - Quality Order Module

Duration: 2 weeks

Tasks:

- Create quality order from completed arrival.
- Load shipment snapshot into Shipment Details tab.
- Load material rows into Container Details and Material Details tabs.
- Implement Open Quality Order button.
- Implement Close Quality Order button.
- Implement Reopen Quality Order for Site Admin only.
- Implement status banner and editability locking.
- Implement material size override with reason and audit.

Acceptance criteria:

- User cannot create quality order before arrival completion.
- User cannot create samples unless quality order is Open.
- Closed quality order is read-only.
- Reopen requires Site Admin and reason.
- Material override creates audit log.

Phase 5 - Sample, Reading, Observation, and Defect Module

Duration: 3 weeks

Tasks:

- Build dynamic sample form based on material group.
- Create sample header fields:
 - carton count.
 - carton identifier.
 - sample scope.
 - sample size.
 - grower.
 - pallet.
 - pack code.
 - date code.
 - lot.

- label.

- Build readings panel from reading type configuration.
- Build observations panel.
- Build dynamic defects panel from `qms\_material\_group\_defect`.
- Implement create, save, edit, soft delete sample.
- Implement defect percentage calculations.
- Implement required fields and threshold validations.

Recommended calculation approach:

- Store raw defect count/value.
- Store calculated percentage per sample.
- Store calculation version.
- Recalculate on sample save.
- Preserve report output using a report snapshot when PDF is generated.

Example defect percentage:

$\text{defect_percentage} = \text{defect_value} / \text{effective_sample_size} * 100$

Where effective_sample_size is:

- sample size if maintained;
- otherwise carton count multiplied by material size, if both available;
- otherwise manually entered effective sample size with reason.

Acceptance criteria:

- Apple defects are configurable, not hardcoded columns.
- Adding a future fruit family does not require changing sample table structure.
- One material can have many samples.
- One sample can have many readings, observations, defects, and images.

Phase 6 - Image Management

Duration: 1-2 weeks

Tasks:

- Implement upload API with content-type and size validation.
- Store original image in file/blob storage.
- Generate thumbnail asynchronously.
- Create image metadata and business link rows.
- Support image categories:
 - Arrival external.
 - Arrival internal.
 - Temperature display.
 - Data logger.
 - Quality inspection.
 - Sample image.
 - Report only.

- Implement reorder images.
- Implement soft delete.
- Implement admin-controlled thumbnail settings for screen and PDF.

Acceptance criteria:

- Arrival images do not leak into unrelated quality orders.
- Inspection images are linked to correct quality order/sample.
- PDF can include both image sections separately.
- Deleted images disappear from UI but remain auditable until retention cleanup.

Phase 7 - PDF Reporting

Duration: 2 weeks

Report sections:

- Header:
 - company name/logo.
 - report title.
 - page number.
 - quality order number.
 - report date.
- Shipment details:
 - BOL, container, vessel, voyage, arrival date, supplier, loading/origin details.
- Material table:
 - material, description, origin, material group, quantity, UoM.
- Sample sections:
 - sample header.
 - readings.
 - major defects.
 - minor defects.
 - observations.
 - acceptance result.
- Image sections:
 - arrival images.
 - quality inspection images.
- Audit footer:
 - generated by.
 - generated at.
 - calculation version.

Tasks:

- Build report data query/stored procedure.
- Build calculation service.
- Store report generation log.
- Store generated PDF archive path.

- Implement report regeneration rules.
- Implement PDF preview and download.

Acceptance criteria:

- Report is reproducible for a closed order.
- Report shows separate image groups.
- Defect rows and readings expand dynamically by material group.
- Report generation does not block user screen for large image sets.

Phase 8 - Admin and Configuration

Duration: 1-2 weeks

Tasks:

- Defect catalog maintenance.
- Material group to defect mapping.
- Reading type configuration.
- Required field configuration by material group.
- Thumbnail screen/PDF settings.
- Role and permission matrix.
- Audit log viewer.
- OData health/check page.

Acceptance criteria:

- Admin can add future fruit defects without schema change.
- Site Admin can reopen closed orders with full audit.
- Thumbnail settings affect UI and PDF output.

Phase 9 - Testing and UAT

Duration: 2 weeks

Test scenarios:

- Container appears under one BOL.
- Container appears under multiple BOLs.
- SAP OData unavailable before arrival save.
- SAP OData unavailable after arrival save.
- Arrival checklist incomplete.
- Arrival completed, create quality order.
- Quality order opened, samples created.
- Multiple samples for one material.
- Material size missing, override with reason.
- Closed order cannot be edited.
- Site Admin reopens order.
- Image upload and thumbnail generation.
- PDF report generation with many samples and images.

- Future fruit material group with different defects.
- Concurrent users creating samples for the same quality order.

Acceptance criteria:

- No orphan data.
- No duplicate sample numbers.
- No wrong BOL/container selection.
- No edit allowed after close unless reopened.
- PDF matches business report requirements.

Phase 10 - Deployment

Duration: 1 week

Tasks:

- Prepare environments: Dev, Test, Production.
- Configure SAP OData destinations and credentials.
- Configure SQL Server connection and migration user.
- Configure file/blob storage.
- Configure background workers.
- Configure backups and retention.
- Configure monitoring and logging.
- Run smoke tests.
- Train key users.
- Go-live with hypercare.

Acceptance criteria:

- Production configuration is documented.
- Backup and restore tested.
- SAP OData connectivity tested.
- User roles tested.
- Rollback plan approved.

10. Recommended Timeline

Week	Workstream
1	Freeze scope, SAP field mapping, DB model approval.
2	SQL Server schema, seed master data, audit framework.
3	SAP OData consumer and container/BOL search.
4	Arrival Overview save, complete, snapshot, images.
5	Arrival checklist validation and arrival report.
6	Quality order header, material rows, status lifecycle.
7	Open/close/reopen rules and material size override.
8	Sample header and dynamic reading model.
9	Defect catalog, material group mapping, sample defects.
10	Image management, thumbnails, reorder, soft delete.
11	PDF report data query and calculation service.
12	PDF layout, image sections, report archive.
13	Admin configuration screens.

Week	Workstream
14	Security, audit viewer, performance tuning.
15	SIT/UAT fixes.
16	Deployment, training, go-live readiness.

11. Field Mapping Corrections from Workbook

Arrival fields

Keep these in arrival checklist or arrival header:

- BOL_NO -> `qms\_arrival.bol\_no`.
- CONTAINER -> `qms\_arrival.container\_no`.
- EBELN -> `qms\_arrival.ebeln`.
- SHIPMENTNO -> should become generated `arrival\_no` or `internal\_shipment\_no`, not the only primary key.
- COMPANYCODE -> `bukrs`.
- CREATED_ON / CREATED_BY -> use datetime2 and user id.
- Seal, cargo, temperature, logger, and photo flags -> checklist columns.

Shipment fields

Move to qms_shipment_snapshot:

- LOADING_DATE, SAILING_DATE, EXAMINATION_DATE, ARRIVAL_DATE, UNLOADING_DATE, INSPECTION_DATE.
- TRANSIT_DAYS, TIME_BAR, LOADING_PORT, LOADING_COUNTRY, ARRIVAL_PLACE.
- VESSEL_NAME, VOYAGE_NUMBER, PULLOUT_DATA, RECEIVING_DATE.
- TIME_BAR_EXCEED, LOGGER_SERIAL, INSPECTION_POINT, JOINT_SURVEY.

Header fields

Move to quality order/material rows:

- QCNO -> `quality\_order\_no`.
- MATERIAL -> `quality\_order\_material.material\_no`.
- MATERIAL_DESC, ORIGIN, CLASS, WEIGHT, SIZE, MAT_GROUP, MAT_GROUP_DESC, MAJOR_CATEGORY -> `qms\_quality\_order\_material` snapshot.
- MENGE / MEINS / WERKS / LGORT / BUKRS -> `qms\_arrival\_item` and copied into report snapshot as needed.

Lines fields

Split as follows:

- GROWER, PALLET, SAMPLE_SIZE, CARTON_COUNT, PACK_CODE, DATE_CODE, LOT, LABEL -> `qms\_sample`.
- BRIX, PUC, PHC, STICKER, FIRMNESS, GROSS_WEIGHT, NET_WEIGHT, TARA, COLOUR, WAXING, DOWNGRADE, PACKAGING_MAT -> `qms\_sample\_reading`.
- WASTE_DECAY, SCALD, BITTER_PIT_PLARA, LENTICELS_BREAKDOWN, CALYX_MOLDS, MECHANICAL_INJURY, STEM_INJURY, SHRIVELLING, SUNBURN, CRACK, WAX_RESIDUE, RED_BLUSH, INSECT_DAMAGE, RUSSETING, CHEMICAL_RESIDUE, BRUISING -> `qms\_sample\_defect` using `qms\_defect\_catalog`.

12. Key Risks and Fixes

Risk	Impact	Fix
Container reused across BOLs	Wrong inspection context.	Mandatory BOL/PO disambiguation.
SAP OData changes after arrival	Inconsistent reports.	Save immutable arrival snapshot.
Defects stored as columns	New fruit requires schema change.	Defect catalog + sample_defect rows.
Readings stored as columns	Future readings require schema change.	Reading type + sample_reading rows.
Closed orders edited	Audit/compliance issue.	Status locks and reopen workflow.
Image ownership unclear	Images appear in wrong report.	Image asset + image link with owner type.
Missing material size	Wrong calculations.	Override workflow with reason and approval.
Concurrent sample creation	Duplicate sample numbers.	DB sequence/transaction and unique index.
Long PDF generation	Poor UX.	Background report job and archive.
OData downtime	Users blocked before arrival.	Clear error before save; SQL snapshot after save.

13. Immediate Next Actions

- Approve the normalized database model.
- Convert the current workbook structure into the proposed tables.
- Confirm OData field names and keys from built CDS views.
- Define first fruit family configuration: Apple.
- Finalize defect threshold formulas.
- Decide sample scope default: one carton or multi-carton.
- Build database scripts first, then OData consumer, then Arrival Overview.
- Avoid building the PDF first; reporting should come after sample/defect data is stable.

14. Final Assessment

The corrected design is strong if treated as a **QMS execution layer after Customer Arrival**, not as a replacement for SAP QM and not as a direct SAP database reader. The highest-value fix is normalization: sample data, readings, observations, and defects must be independent child entities. That design lets the system support apples now and other fruit families later without redesigning the database.

The most important technical rule is this:

Before Arrival save, read from SAP CDS/OData. After Arrival save, read QMS data and locked SAP snapshot from SQL Server.

This protects auditability, performance, report consistency, and user trust.